

The Economics of Recursive Self-Improvement

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June 18, 2026

PRELIMINARY AND INCOMPLETE. Please do not circulate but we welcome comments.

Content: opinionated take on existing RSI theory & data.

Audience: (1) economists who want to get up to speed on RSI theory and data; (2) AI researchers who want to follow econ arguments, esp. about what data is useful.

Timeline: June 5 send for comments; June 11 circulate publicly.

Organization: Aim for ≤ 15 pages in the body; appendices can be long.

Abstract

This note offers a short and opinionated review of the economics of recursive self-improvement (RSI). First, we build a sequence of progressively more complex models of AI progress to highlight the key feedback loops behind RSI. We extend existing theoretical work to distinguish between “narrow” and “broad” AI capabilities, capturing the possibility that capabilities first accelerate narrowly in verifiable domains before expanding to less verifiable ones. Second, we document existing measures of key parameters needed to calibrate our model of RSI, and provide a wish list of empirical objects that AI companies can measure and share publicly. Based on these data, we discuss the plausibility of a capabilities acceleration due to RSI and its possible economic impacts.

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Acknowledgements: Thanks for comments from Anson Ho and Thomas Houlden.

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Support: We gratefully acknowledge administrative and financial support from METR, and hosting from Constellation.

1 INTRODUCTION (LA, BH)

Motivation. The continuous improvement of technology is the cornerstone of modern economic growth. Each industrial revolution has been fueled not by a single technological breakthrough, but by a virtuous cycle of thinkers and tinkerers improving technologies iteratively (Mokyr, 2002). For the first time, a general-purpose technology may learn to do the thinking and tinkering itself: artificial intelligence.

Core argument. The critical argument is that we should expect an *acceleration* in the growth of AI capabilities, relative to trend, as the models become good enough to help with AI research. The degree of acceleration depends on the core feedback loop: for each unit increase in model capabilities, how much does that increase the capabilities of the next generation of models? This note tries to organize theory and data to help address that question, and so help predict the degree of acceleration to expect.

Defining RSI. The term “recursive self-improvement” (RSI) has been used in a variety of ways, often defined as a technology contributing to its own improvement, which could be interpreted as applying to almost any technology.¹ We use RSI to refer to the general area, but most of our discussion uses a set of less-ambiguous terms defined below.

Term	Definition
Technological feedback loops	When higher capability levels increase future capability levels, holding other inputs fixed (human R&D, capital, etc.). Arguably most technologies exhibit feedback effects, and AI has certainly exhibited a degree of feedback for a long time.
R&D automatability	When AI is able to autonomously make technological improvements at least equivalent to those made by unaided human researchers, at an equal cost. Even when this is true, humans could still be employed in R&D due to slow diffusion, regulation, or complementarities.
Self-sustaining growth	When AI systems sustain accelerating progress across capability thresholds without any growth in exogenous inputs (human labor, training compute, etc.). ²
Capabilities acceleration	An increase in the historical rate of progress of capabilities.

¹The term was coined by Yudkowsky (2007), extending an argument by Good (1965). See a survey of related definitions in <https://tecunningham.github.io/posts/2026-06-05-rsi-definitions.html>.

²This is related to definitions in the literature of “explosive”, “hyperbolic”, and “superexponential” growth. In models with a cardinal capability metric this is often formalized as exponential or unbounded growth. Since some capability metrics are interval scales, we generally use a threshold-crossing version: whether systems traverse increasingly demanding capability levels at an accelerating pace without growth in exogenous inputs.

Defining AI capabilities. A difficulty in discussing RSI is that there is no canonical metric of an AI model’s capability or intelligence. We will often refer to model “capability” without specifying which metric we mean. Some common candidates are ECI (Epoch Capability Index, Epoch AI (2025), roughly an average of benchmark scores), time horizon (Kwa et al., 2025), base-model perplexity, and “effective compute” (the amount of training compute required to reach an equivalent capability level for some fixed architecture). Some important notes about capability metrics are: (1) most metrics are highly correlated across models (Ruan et al., 2024), which is reassuring that a scalar capability metric can be used without too much loss; (2) model capabilities are often described as “jagged”, meaning they have a very different correlation structure to human capabilities; (3) some common capability metrics are “interval scales” meaning that we cannot meaningfully talk about proportional rates of growth, though we can talk about acceleration across equal intervals (e.g. ECI);

Modeling RSI. We construct a sequence of progressively richer models of RSI. We do so largely informally through causal loop diagrams, rather than emphasizing equations, though we typically supplement the diagrams with equations.

The models all collapse down to a single feedback loop where stronger AI capabilities increase effective R&D, which in turn produces further AI capabilities. Our basic model distinguishes three different production functions, which combine to form this feedback loop: the production of AI capabilities, algorithmic efficiency, and R&D inputs. A fourth production function, for output, determines economic impacts.

We then extend the model to allow for bottlenecks, where the necessity of humans may break the feedback loop; for economic feedback loops, where higher output finances further compute; for experimental compute; and for data.

Finally, we distinguish narrow from broad capabilities, allowing RSI affect easier-to-verify tasks (coding, math, benchmark optimization) without spilling over into harder-to-verify tasks.

Existing empirical evidence & requests for data. Our models point to a set of critical measurable parameters. We review existing empirical measures of these parameters and put forward a list of measures that would be useful for AI companies to provide. The most useful additional data would be (i) the growth rate of algorithmic efficiency in labs, (ii) lab expenditures across inputs (human R&D, data, experiment compute, inference for R&D).

Plausibility and implications. In taking stock of the theory and evidence, we also discuss the plausibility and impacts of an acceleration in capabilities. The strongest evidence in favor of an acceleration is the combination of high estimated returns to research, relative to other types of R&D (Bloom et al., 2020), and rapidly advancing AI capabilities on coding and research benchmarks. The strongest evidence against is that some AI R&D benchmarks remain far from saturated, algorithmic progress has historically been tied to compute scaling, and continued compute scaling may itself become constrained by GDP or power constraints. Political and social barriers are likely to slow deployment even if the technical conditions for self-sustaining capability acceleration are met.

Position in the literature. The economic literature on RSI is founded on Aghion et al. (2017), who derived conditions under which AI-driven automation could generate accelerating or even explosive growth, and Davidson (2023), whose compute-centric framework offered an early formal model of AI takeoff dynamics. We complement an emerging economics literature on RSI in

three ways. First, Davidson et al. (2026) formally develop a semi-endogenous growth model with an innovation network, and Eth and Davidson (2025) calibrate a model of a software-only takeoff. Our framework is deliberately simple and often informal, allowing us to be more wide-ranging. Second, following Cunningham and Shetty (2026), we emphasize the distinction between narrow and broad capabilities in both theory and the discussion of economic impact. This distinction is largely set aside in the review articles of Trammell and Korinek (2025) and Jones (2026). Third, we provide a wish list of empirical objects that AI companies can both measure and feasibly share openly that would be informative for the public conversation on RSI. This supplements the existing empirical efforts that we review (e.g. Whitfill and Wu 2025, Gundlach et al. 2025, Epoch 2026).

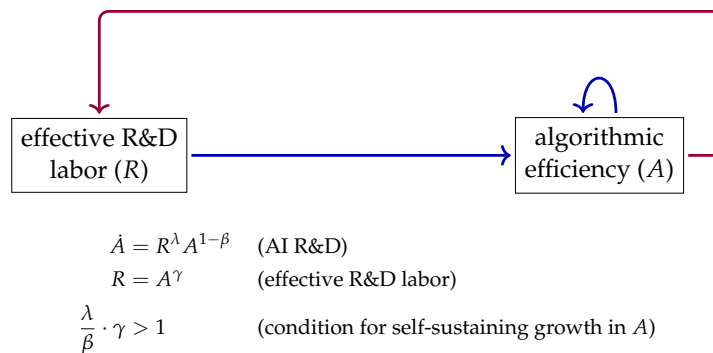
2 A SIMPLE MODEL OF RECURSIVE SELF-IMPROVEMENT (AK,BH,TC)

In this section we present a sequence of models illustrated with causal loop intuition and readability, which can be read as specifying a system of functions. In most cases we additionally propose some simple functional forms. It is important to note that the iso.

Because many empirical measures of AI capability are interval scales, we write capabilities as an interval-valued index, C . Thus we avoid claims about proportional growth in capabilities, and instead state conditions under which systems cross equal capability intervals at an accelerating rate.

In the illustrative equations below, algorithmic efficiency A , compute, labor, and output are cardinal quantities, but capability C is defined only up to positive affine transformation. A convenient benchmark specification is $C = \theta \log(AT)$, so that proportional increases in effective compute correspond to additive increases in the capability index. Effects of capability on effective R&D labor are therefore written as semi-elasticities, e.g. $R = L \exp(\gamma C)$, rather than as elasticities with respect to a cardinal capability stock.

The core feedback loop. All our models are built around a simple feedback loop. We assume the growth in AI algorithmic efficiency (A) is a function of R&D input (R) and the existing level of efficiency (“ideas getting harder to find”, Jones 1995). We add the assumption that algorithmic progress increases *effective* R&D labor (remaining agnostic whether a substitute or complement to humans).

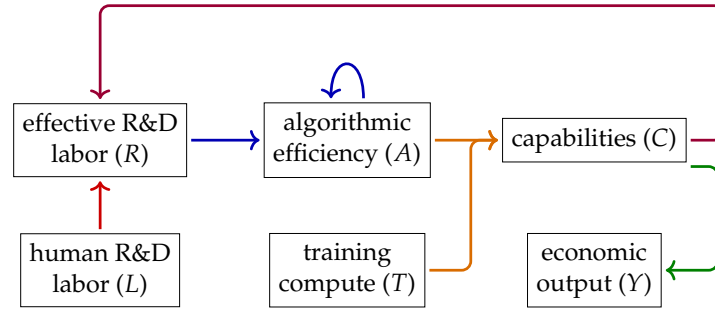


The self-sustaining growth condition will be satisfied when a 1% increase in algorithmic efficiency causes a sufficient increase in effective R&D labor to increase algorithmic efficiency of the next generation by 1% or more. We expect meaningful R&D acceleration only with sufficiently advanced AI, meaning γ will vary, but our condition is sufficient for local acceleration.

PW, CW: don't understand this last sentence. Why is it not constant?

Our basic model. We extend the core feedback loop into four functions, indicated with differently colored arrows:

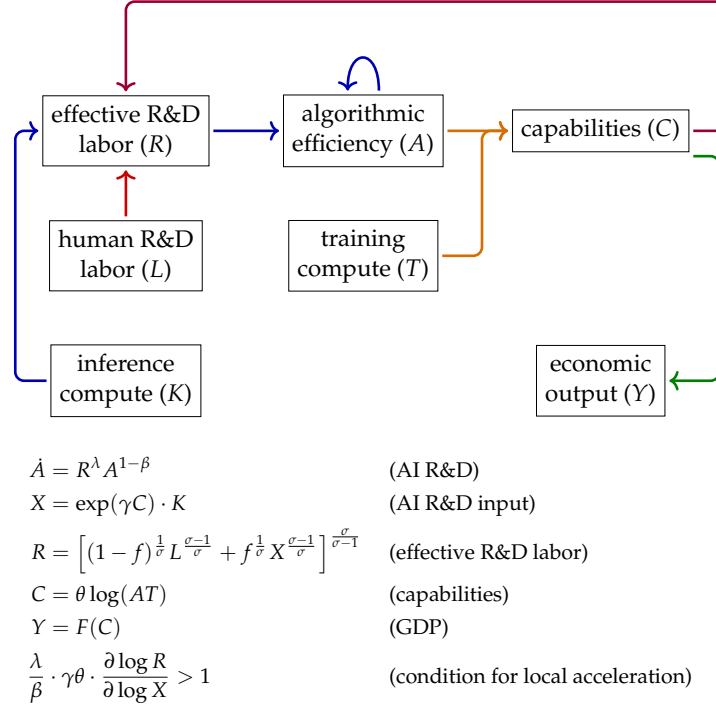
1. **Economic output** is determined by capabilities.
2. **AI capabilities** are determined by pretraining compute and algorithmic efficiency. We represent them with an interval-valued index, C , rather than a cardinal stock.
3. **Algorithmic efficiency** is improved by R&D labor, as above.
4. **Effective R&D labor** is a function of **human R&D labor**, augmented or automated by AI capabilities.



$$\begin{aligned}
 \dot{A} &= R^\lambda A^{1-\beta} && \text{(AI R\&D)} \\
 R &= L \cdot \exp(\gamma C) && \text{(effective R\&D labor)} \\
 C &= \theta \log(AT) && \text{(capabilities)} \\
 Y &= F(C) && \text{(GDP)} \\
 \frac{\lambda}{\beta} \cdot \gamma \theta &> 1 && \text{(condition for acceleration)}
 \end{aligned}$$

The critical condition is nearly identical, except that the feedback from algorithmic efficiency to effective R&D labor now passes through the interval capability scale. The condition should be interpreted as one for accelerating progress across equal capability intervals, not for exponential growth in a cardinal capability stock.

Bottlenecks. If AI cannot entirely replace human researchers then the degree of substitability becomes important. We allow effective R&D labor, R , to depend on inference compute, K ; the value of inference compute is augmented by capabilities, C . Some fraction f of effective labor is automated by AI, with an elasticity of substitution σ between humans and AI. If σ is below 1, then labor bottlenecks algorithmic efficiency (Aghion et al., 2017).

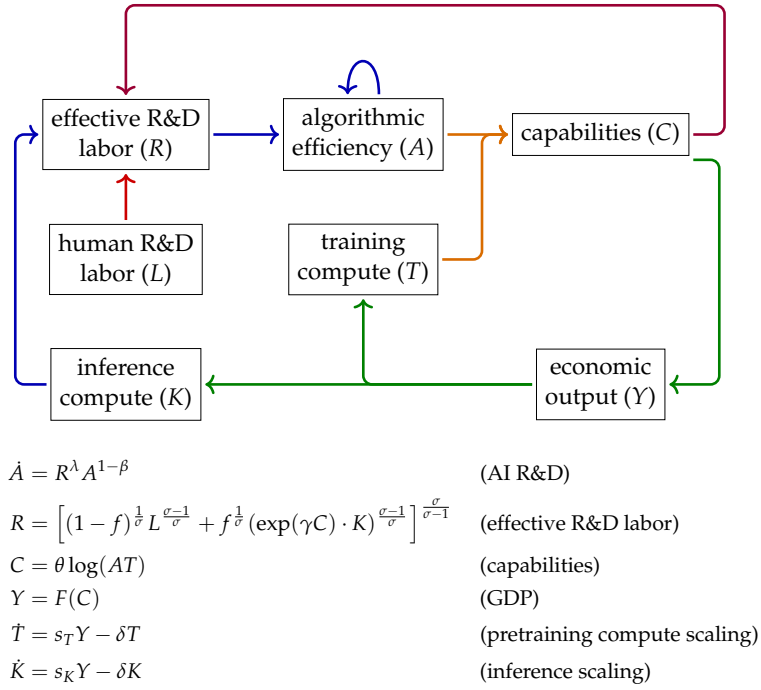


If inference compute increases disproportionately relative to human labor, bottlenecks reduce the marginal value of AI in research to zero. In terms of the causal loop diagram, this would remove the two arrows from capabilities C and inference compute K to effective R&D labor R , thus breaking the feedback loop and preventing self-sustaining acceleration.

In the math, the disappearance of the $C \rightarrow R$ arrow and $K \rightarrow R$ arrow correspond to the *elasticity of effective R&D labor to AI inputs* going to zero. This highlights the critical role of this important elasticity (Davidson et al., 2026), which we return to in the measurement section. We note that, under cost minimization, this elasticity is locally equal to the inference compute share of the R&D budget, which can be measured over time.

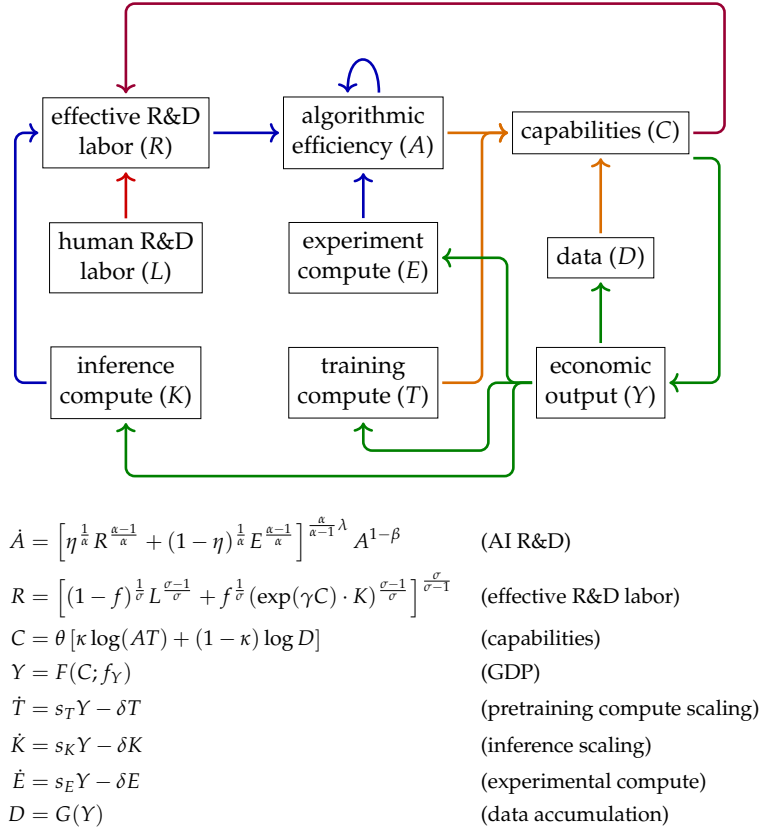
Economic feedback loops. The previous model implicitly held constant the level of compute available for training and as R&D labor. This isolated the role of algorithmic efficiency progress in driving a software-only singularity. There are also **economic feedback loops**: higher AI capabilities driving higher GDP, which in turn finances further investment in compute and R&D, further advancing AI capabilities (Davidson et al., 2026).

This feedback loop is necessary if advances in AI capabilities require continued scaling of compute. On the other hand, if advances in capabilities can be developed without scaling compute – e.g. by developing more sample-efficient algorithms – then this feedback loop may not be necessary and a “software-only singularity” is possible.



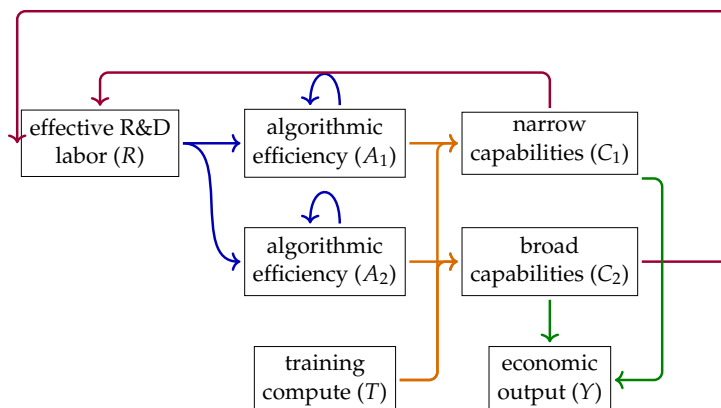
An extended model: experimental compute and data. The figure presents an extended model – centered around the core R&D $\leftrightarrow$ capabilities feedback loop – to both clarify other forces and also to help interpret what we see in the data. Two additional forces have been added:

1. **High-quality data** is a critical input to the AI capabilities production function. Unlike narrow domains that have seen spectacular progress via synthetic data produced by inference compute, lack of real-world data may bottleneck broad RSI (Millidge, 2025) and require learning-by-doing via real-world deployment (Farboodi et al., 2025; Erdil et al., 2025a).
2. **Experimental compute** may be a bottleneck to algorithmic progress. Advances in algorithmic efficiency may require researchers to tinker, which in turn requires compute for experiments. Whitfill and Wu (2025) estimate that experimental compute is a strong bottleneck to algorithmic progress.



Narrow vs. broad capabilities. One of our distinctive claims is the importance of differentiating between progress in **narrow capabilities** versus in **broad capabilities**.

This distinction can be made in several ways: tasks with output which is “verifiable”, versus those in which it is not; tasks which are not “messy”, versus those which are; those not requiring “taste”, versus those which do.



Although the preceding diagrams abstracted away from this distinction, we think it will be crucial. There is suggestive, growing evidence that advances in capabilities are occurring more quickly in narrow domains (Cunningham and Shetty, 2026). If AI R&D has differential impacts on the progress of narrow and broad capabilities, this might bottleneck improvements in algorithmic efficiency. For example, the lack of ‘research taste’ in AI systems might prevent compute from fully substituting for researcher labor, which could delay or even preclude rapid progress in broad capabilities. Yet, we also emphasize that narrow and broad lie on a spectrum. This means that what starts out as ‘merely’ a narrow capability acceleration might eventually drive broader improvements as capabilities advance step-by-step up the narrow→broad ladder (Farboodi et al., 2025): improvements in performance on ‘routine tasks’ could boost the pace of algorithmic improvements which, in turn, drive improvements in performance on ‘substantial tasks’ which, by further boosting algorithmic improvements, drive improvements in performance on ‘open-ended problems’.

Economic impacts. Recursive self-improvement may have large downstream economic impacts on GDP – but it need not. In the diagrams above, this possible impact is represented by the arrows from AI capabilities to economic output Y . It is helpful to distinguish four cases:

	Narrow RSI	Broad RSI
Small econ impact	AI capabilities are astounding as measured by benchmarks containing cheap-to-verify tasks, but the wider economy is bottlenecked by messier hard-to-verify tasks	Unlikely, but could occur because of regulation
Large econ impact	AI capabilities as measured by benchmarks with cheap-to-verify tasks are representative of true economic capabilities	Economically transformative

Interpreting interval-scale critical conditions. Since C is an interval scale, C and $a + bC$ with $b > 0$ represent the same capability facts. This means that claims about the growth rate $\dot{C}/C$ or about “doubling capabilities” are not invariant to admissible transformations of the scale. In particular, exponential growth in C is not a well-defined primitive property of the world: after the transformation $C' = a + bC$, $\dot{C} = rC$ becomes $\dot{C}' = r(C' - a)$, and the apparent proportional growth rate changes.

The most conservative response is to write the model in terms of threshold-crossing times rather than proportional growth in capabilities. Suppose, holding other inputs fixed, that capability progress can be summarized by

$$\dot{C} = M(C)R(C)^\lambda,$$

where $M(C)$ is the amount of capability progress produced by a unit of R&D input at capability level C and $R(C)$ is effective R&D labor supplied by systems at that capability level. Then the local analogue of the critical condition is not an elasticity condition, but a semi-elasticity condition:

$$\frac{d \log \dot{C}}{dC} = \frac{d \log M(C)}{dC} + \lambda \frac{d \log R(C)}{dC} > 0.$$

This condition is invariant to affine rescalings of C up to multiplication by the positive constant $1/b$, so its sign is invariant. It says that the speed at which systems cross equal capability intervals is itself increasing: the boost to effective R&D labor from moving one interval unit up the capability scale outweighs the increase in research difficulty over that interval. If the left-hand side is bounded below by a positive constant over a long range, then the speed of threshold-crossing rises at least exponentially with the interval index, and the time needed to traverse all further finite thresholds,

$$\int_{C_0}^{\infty} \frac{dC}{M(C)R(C)^\lambda},$$

can be finite. If it is negative, progress on the interval scale decelerates even if some cardinal transformation of C could be chosen to make another quantity look exponential.

This nests the critical condition in our basic model as a special case. Holding training compute fixed, $C = \theta \log A$. With effective R&D labor exponential in the interval score, $R = L \exp(\gamma C)$, and $\dot{A} = R^\lambda A^{1-\beta}$, we have

$$\dot{C} = \theta \frac{\dot{A}}{A} = \theta L^\lambda \exp \left[\left(\lambda \gamma - \frac{\beta}{\theta} \right) C \right].$$

Thus interval-scale progress accelerates iff

$$\lambda \gamma > \frac{\beta}{\theta} \iff \frac{\lambda}{\beta} \gamma \theta > 1.$$

In this interpretation γ is not an elasticity of R&D labor with respect to a cardinal capability stock. It is a semi-elasticity: the percent increase in effective R&D labor produced by one additional interval unit of capability. Likewise, θ converts log algorithmic efficiency into interval capability units. Under $C' = a + bC$, γ is divided by b and θ is multiplied by b , so the product $\gamma \theta$ and the critical condition are unchanged.

There are several defensible modeling choices here. One can choose a cardinal operational target, such as effective compute, labor-equivalent research output, willingness-to-pay, or odds of

success on a family of tasks, and then apply the elasticity condition to that target. This is often empirically convenient, but the cardinalization is an additional structural assumption. Alternatively, one can stay on the interval scale and estimate $\frac{d \log R}{dC}$ directly: for example, if C is a task-difficulty frontier and research tasks have weights $w(x)$ and success probabilities $p(C - x)$, then $R(C)$ might be proportional to $\int w(x)p(C - x) dx$, making the critical question whether enough valuable R&D tasks lie near the current frontier. Finally, one can avoid a single scalar capability measure altogether and model a vector of task-domain capabilities. The critical condition then becomes a spectral-radius condition for the Jacobian of the feedback map from capabilities to effective R&D and back to capabilities. Across these choices, the invariant object is not “exponential growth in capabilities” but whether the system crosses economically or scientifically meaningful capability thresholds at an accelerating rate, and whether that acceleration is strong enough to overcome increasing research difficulty.

Important extensions. We suggest two further extensions are important to pursue.

(1) *Scale-dependent algorithmic progress.* Gundlach et al. (2025) document that the most valuable algorithmic improvements to date have increased measured AI capabilities by an amount which itself increases with the compute used in training. For example, when you use only 10^{12} FLOPs of training compute, moving from LSTMs to the transformer architecture produces a smaller capability gain than when you use 10^{23} FLOPs. The theoretical literature has not formalized this idea or its implications. We offer a summary in Appendix A.

(2) *Endogenous automation.* A key question for forecasting AI impacts is understanding the possibility of bottlenecks. As discussed in the subsection on bottlenecks, if automation advances sufficiently quickly, then bottlenecks will not bite – even if humans and AI are highly complementary. Thus, a theory explaining the evolution of automation is important: how will the fraction of research tasks performed by AI rather than labor, f , grow over time?

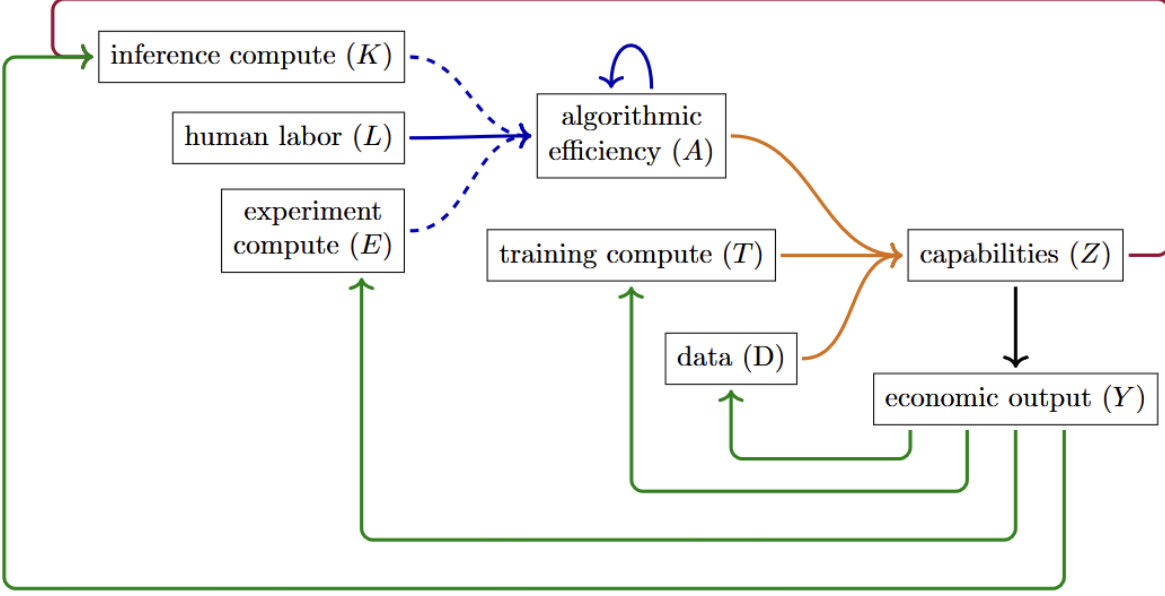
3 DATA RELEVANT TO RSI (TC,PW)

In this section, we first highlight the most important overall parameter to estimate and sketch how to estimate it. Then, we discuss supplementary measures grouped around the extended model – repeated below for reference – with its four stages: (1) R&D inputs; (2) model training; (3) R&D acceleration; (4) the effect of AI on GDP. Each of the four subsections below covers one stage of the model, and each row below refers to one object in that stage.

3.1 Measuring the Explosion Condition. In the bottlenecks section, we derive a local explosion condition: capabilities growth is hyperbolic/explosive when the left-hand side below exceeds 1, and exponential when it falls below 1.

$$\frac{\lambda}{\beta} \cdot \gamma \theta \cdot \frac{\partial \log R}{\partial \log Z^{\gamma} K} \gtrless 1 \quad (1)$$

Although we derive this condition within a specific model, we believe it holds much more generally. The CES form of the research-input function is not essential: research input can be any constant-returns-to-scale function of arbitrary inputs, provided effective AI labor $Z^{\gamma} K$ is one of them. The condition is attractive because each factor maps to an observable quantity:



- the returns to R&D, $\frac{\lambda}{\beta}$. In steady state $\frac{\lambda}{\beta} = \frac{g_A}{\text{elasticity weighted avg growth of all research inputs}}$, so it can be estimated as the growth rate of algorithmic progress divided by the growth rate of research inputs. In the simpler models, research inputs are just labor. In more complicated models, research inputs include more things in which case we need to do a weighted average. Note that this approximation only works in the case where g_A is constant.
- The AI quality-quantity tradeoff γ , recovered by comparing the returns to scaling training compute with the returns to scaling inference compute.
- The elasticity of capability to effective algorithm, θ , recovered by measuring the return to benchmark-derived capabilities, such as ECI, when scaling training compute, holding algorithmic efficiency constant.
- the elasticity of effective research input to effective AI labor, $\frac{\partial \log R}{\partial \log Z \gamma K} = S_k$, the share of research spending devoted to inference. Because a cost-minimizing firm sets each input's spend share equal to its output elasticity, this term is directly observable. Unlike the other two, it moves over time: it was near zero before 2024 and has risen quickly since.

In our view, estimating this product, tracking it over time, and extrapolating when it might cross 1 is the single most important empirical exercise for assessing whether a capabilities explosion will occur. The next two sections discuss ways to estimate each of these parameters. After that, we discuss supplementary data that helps build a broader picture of AI R&D automation.

3.2 Data on R&D Inputs. This subsection covers the inputs to algorithmic progress, $\dot{A} = \phi(R, E) A^{1-\beta}$, where effective R&D labor R combines human labor L and inference compute K , and E is experiment compute. We collect data on each input, on their growth rates, on how they substitute, and on their budget shares. The rows we find most useful are those mapping to the three terms of the explosion condition (Section 3.1): the returns to R&D $\lambda/\beta = g_A/g_R$, the elasticity γ , and the inference spend share S_k .

Question	Existing Data	Potential for New Data
Data on AI R&D inference compute (K) and its share of total R&D spend (S_k).	We have almost no data on lab expenditure on inference or its share relative to other R&D inputs.	This data would likely have to come from labs and ideally would be tracked over time.
Data on human researchers (L).	Epoch AI (2026) estimates growth rates of around 2–3X/year in total staff for frontier labs over 2023–2025. Whitfill and Wu (2025) estimate headcount growth rates of 2–3X/year for Anthropic and OpenAI over 2022–2024. Cottier et al. (2024) estimates that around 1/3 of the cost of training models was spent on researcher salaries.	It would be useful to have granular data from labs on staff disaggregated across seniority and across researchers vs engineers, etc.
Data on experiment compute (E) and its growth g_E.	You (2025) estimates that in 2024 only 10% of OpenAI’s R&D compute was on final training runs, while 90% was on experimental compute (i.e. training which did not directly affect weights used in a public product). Denain and Wu (2026) estimate similar ratios for Chinese companies. Morrison et al. (2026) estimate similar ratios for OLMo 3 family.	Labs have rich data on expenditure on experiment compute across domains and across time; this would be valuable for estimating the degree to which automated research will be bottlenecked by experiment compute.

Question	Existing Data	Potential for New Data
Data on substitutability of labor and experiment compute (σ).	Whitfill and Wu (2025) try to estimate the substitutability between labor and experiment compute in AI R&D, using variation in relative prices, but their findings are inconclusive.	It is challenging to run randomized experiments that vary R&D labor and experimental compute over reasonable ranges. However it would be valuable to run surveys asking team managers about their expectations. A basic survey would ask managers how much more progress they would make toward their goal (e.g., compute efficiency) in the following scenarios: (A) twice as many researchers (compute fixed); (B) twice as much compute (researchers fixed); (C) twice as many researchers and twice as much compute. The answers to these questions would pin down the degree of substitutability between researchers and experimental compute. Second, a cheaper version of a randomized experiment could use AI labor instead of human labor: given an ML problem, vary the amount of AI labor and GPUs available for experiments, then measure how much progress is made.
Tradeoff between training and inference compute (γ).	The parameter γ governs the returns to scaling capabilities Z versus scaling inference compute K , since we model effective AI labor as $Z^\gamma K$. The Epoch GATE model (Erdil et al., 2025b) offers one estimate of γ .	Public benchmarks have compared the returns to scaling training compute versus the returns to more inference tokens.

3.3 Data on Model Training. Data on model training is useful for two separate reasons: (1) it helps identify historical growth in algorithmic efficiency; and (2) it helps predict the impact of future growth in algorithmic efficiency.

Question	Existing Data	Potential for New Data
Algorithmic efficiency (A) and its growth (g_A).	The level of algorithmic efficiency is inferred from model training data, by comparing the compute required to reach a fixed level of capability over time (Ho et al., 2024). Ho et al. (2024) estimate growth of approximately 3X/year in algorithmic progress for pretraining, and Whitfill et al. (2025) estimate 3.5X/year for the entire stack. Ho (2026) estimates 10X/year for the entire stack. Gundlach et al. (2025) instead argue that algorithmic progress has grown at different rates across scales: 1.5X/year at small scale, and 2.5X/year at frontier scale, for pretraining.	Data on algorithmic efficiency since 2024 is very weak due to limited compute estimates for newer models. Additionally raw algorithmic efficiency becomes a more difficult metric in post-training, where compute can be spread across different tasks.
Overall model training laws.	The basic model remains Chinchilla (Hoffmann et al., 2022), which maps data and compute to pretraining loss.	It would be useful to summarize the latest knowledge from public sources on scaling laws, to help calibrate the RSI model using the substitutability of different inputs in model training. Frontier AI labs maintain detailed estimates of scaling laws, which would be even more useful.
Growth in training compute.	Training compute has been growing 4–5X a year since 2010 (Rahman and Owen, 2024). Whitfill et al. (2025) discuss possible GDP constraints on continued compute scaling.	There is scarce data on model training compute since 2024.
Growth in data.	There are minimal existing estimates of data spend by labs.³	Researchers could recover estimates from labs or partial estimates from consulting Surge AI, Scale AI, or other third-party data providers.

3.4 Data on R&D Acceleration. Until about a year ago there was very little evidence for LLMs meaningfully accelerating the pace of R&D. We now have some data but it is very difficult to quantify.

Question	Existing Data	Potential for New Data
The effect of AI use on researcher productivity.	We have some self-reported productivity acceleration results from surveys: 2X from METR’s survey of developers (Becker 2026); 4X from the Mythos system card. The magnitudes are very difficult to interpret, but there is a clear dramatic upward trend. Favaro and Clark (2026) report lines of code merged per engineer per day rose roughly 8X between 2024 and Q2 2026.	An experiment on AI researchers would be extremely useful; however there are many practical difficulties in running and interpreting such experiments (see discussion in Becker et al., 2026). Deeper surveys would also be very useful, such as asking managers in-depth questions about their best guess at the causal effect of AI adoption.
The strength of autonomous R&D.	AI agents have recently been able to advance the frontier on some optimization problems; however it is difficult to find a scale to measure their overall ability.⁴ Recent system cards report mixed results on benchmarks of AI R&D, generally outperforming humans on optimization problems when the humans are given 8 or 40 hours, but showing low pass rates on complex debugging problems (Mythos system card, table 2.3.3.A; GPT-5.5 system card, section 9.1.3). These results are difficult to place on a common scale.	It would be very useful to know the performance of autonomous AI systems on real LLM training problems. The ideal data would characterize the returns to expenditure on agentic, human, and hybrid labor.

Question	Existing Data	Potential for New Data
Qualitative reports on capabilities of AI R&D.	It remains unclear how to describe AI's weaknesses in AI R&D relative to human researchers. A common split is between idea generation (taste) and idea execution. Si et al. (2025) find that AI-generated AI research ideas are rated highly relative to human-generated ones. Trehan and Chopra (2026) finds a range of weaknesses, including both taste and execution. The Mythos system card has an extended discussion of the qualitative strengths and weaknesses of Mythos' abilities at AI R&D.	Unstructured interviews with lab staff.

3.5 Data on Model Economics.

Question	Existing Data	Potential for New Data
How much economic value added do OpenAI, Anthropic, and Google DeepMind capture?	We can use firm revenue estimates, but these are likely a small share of value created. Brynjolfsson et al. (2026) use online choice experiments to elicit willingness-to-accept for giving up generative AI chatbots, estimating that consumer surplus substantially exceeds firms' revenues.	How much has this varied over time?

4 DISCUSSION (AR,BJ,CW,LA)

In this section, we assess the evidence in favor of and against explosive capabilities growth.⁵

⁵We do not separately frame the question as whether explosive growth is possible with fixed human inputs. For the relevant notion of explosive growth, continued growth in human research labor cannot be a central driver. If a capabilities explosion occurs by some date t , then the number of human researchers available by that date is finite. Thus, any path to explosive growth that relies on growing human input up to t must also be consistent with some finite upper bound on human inputs.

4.1 How plausible is explosive capabilities growth? In section , we illustrate the paucity of credible empirical estimates for key parameters to evaluate the plausibility of RSI. Thus, we do not attempt a precise calibration of the models. Instead, we discuss what we see as the most important points of evidence for and against explosive capabilities growth.

Best evidence in favor of explosive capabilities growth:

- The Mythos system card showed a 4x researcher uplift from AI use (self-reported) and a 40-hour time horizon at which Mythos beat human researchers.
- Experiment compute accounts for only 10% of total R&D. That implies automated research so far should not be bottlenecked by experiment compute.

Best evidence against explosive capabilities growth:

- AI R&D benchmarks are still far from saturated (see the GPT-5.5 system card) and forecasts for acceleration are mixed (METR, 2025).
- Gundlach et al. (2025) argue that most algorithmic progress thus far has been scale-biased, i.e. returns have been increasing in compute, and have come from big breakthroughs such as the transformer. This suggests that scaling compute and “out-of-distribution” innovation will continue to be important.
- Yet compute bottlenecks could occur if productivity growth is not enough for continued AI capex, that is, if we “run out of GDP” (Halperin, forthcoming). Inelastic power markets could also limit continued scaling.
- The apple-picking model of AI R&D (Cunningham and Shetty, 2026): Terry Tao and Andrej Karpathy report that while models are useful at picking low-hanging research tasks, they quickly lose effectiveness on harder problems, suggesting diminishing returns — though this evidence remains early.

Important evidence that is ambiguous:

- Millidge (2025) argues that “most algorithmic progress is data progress”, from better filtering of pre-training data to the substantial resources spent on post-training data. If true, this reduces the plausibility of explosive capabilities growth to the extent that data collection is difficult to automate. Pre-training filtering seems to be a one-time boost, while post-training data might be more elastically supplied and amenable to automation. We are only beginning to learn how to build better RL environments, and AI systems could arguably help design future ones. And just as chain-of-thought (arguably an algorithmic innovation) was complementary to post-training data, future algorithmic improvements could be as well.
- Mythos system card: on the one hand, Anthropic reports that the key algorithmic advances behind Mythos did not originate from Claude. On the other hand, Claude wrote almost all of the code behind Mythos and researchers report a 4x uplift. This could be interpreted as evidence of high g_f and large f . But if labor remained essential and hard to replace in the most important steps, σ could be low.

Narrow capabilities explosion may come before broad capabilities explosion. It seems plausible that AI R&D agents are effective at optimizing benchmarks, but are less effective at developing major breakthroughs that allow models to learn efficiently in new environments. Nonetheless, spillovers from narrow to broad progress remain possible. For example, an advanced narrow system, in pursuit of benchmark optimization, may discover new capabilities, such as highly sample-efficient algorithms able to update their weights during deployment, that quickly turn narrow progress into broader progress.

Political and social barriers. Even if explosive growth is technically plausible, political and social barriers could stop it in its tracks. Recent policy reactions to Mythos, despite an earlier laissez-faire stance, make us put more weight on these barriers. If compute scaling remains important, then even minor frictions to model deployment will slow down revenue growth and further scaling. Nuclear power, Concorde, and high-speed rail across the United States are all cautionary tales that what is feasible may not be realized in practice.

Forecasts of progress. Although forecasts of AI progress exist, there is no consensus on which metrics to use.⁶ Some notable recent forecasts include:

- In August 2025, experts and superforecasters (METR, 2025)⁷ predicted an 8–20% chance that the growth rate of effective compute would triple by 2029.
- In April 2026, Ajeya Cotra predicted that AI automation of AI research will reach “parity” with human research in early 2030 (Cotra, 2026).
- In May 2026, Jack Clark predicted a 60% chance that by the end of 2028 there will be “an AI system powerful enough that it could autonomously build its own successor” (Clark, 2026).

4.2 Impacts of explosive capabilities growth on the world. Explosive capabilities growth differs from generic AI progress in *speed*: it could compress generations of progress into short periods of time. We highlight several aspects of explosive growth that distinguish it from gradual AI progress, and that may be especially important for understanding its social, economic, and political consequences.

- **Transition costs:** It seems plausible that transition costs increase with the speed of technological change. Labor markets are one such case: the costs of structural transformation may rise with its speed, as many economists argue was the case with the China Shock.
- **Institutional inertia:** In the world of explosive capabilities growth, institutions broadly defined — regulatory capacity, democratic procedures, legal frameworks, infrastructure, and so on — might lag behind (Bostrom, 2014; Ord, 2020). Rapid technological change might also grant political elites a temporary spike in *de facto* power which they could use to capture institutions (Acemoglu and Robinson, 2008).
- **Asymmetric progress across domains:** As noted, we see an explosion in narrow capabilities as more likely in the near term than an explosion in broad capabilities. The former would

⁶Polymarket tracks scores on FrontierMath, Humanity’s Last Exam, and Chatbot Arena.

⁷“Experts and superforecasters both find a three-fold acceleration of effective compute scale-up by 2029 plausible but unlikely, with experts giving it a median 20% chance and superforecasters 8%.”

likely be concentrated in tasks that are easy to verify in closed loops such as software development, games, mathematics, and related areas. This could create acute asymmetries in progress across domains. For example, we may find solutions to long-standing puzzles in mathematics and breakthroughs in computational chemistry that enable new materials, alongside sluggish progress in less self-contained or messier domains.

- **Power imbalances across firms and nation-states:** Even model providers with relatively small gaps in technical skill could see vast gaps open up in effective capabilities and power over the world. Imagine Anthropic or the US government having access to nuclear-level capabilities vastly superior to any other actor.
- **Greater misalignment risk:** Misaligned AI might also create bigger risks under explosive capabilities growth. Sudden increases in model capabilities would provide less time for technical and social safeguards to be put in place.

4.3 Future Steps. For economists, there is a rich set of urgent theoretical, empirical, and policy questions to address.

- With respect to theory, a first priority is to better model the AI research production function. Though we have extended the model to include potential bottlenecks arising from complementarity between labor and compute, there may be further extensions. The production function could go beyond a simple CES. A second priority is to understand how AI capabilities develop with algorithmic progress and other inputs. We have already extended this in one direction by distinguishing narrow and broad capabilities driven by algorithmic progress. There are other directions to extend the model. For example, if algorithmic progress is scale-dependent, as suggested by Gundlach et al. (2025), then the relationship of $Z = A \times T$ may miss an important complementarity between taste and compute (see also Appendix A).
- With respect to empirics, many of the most important model parameters are currently weakly identified and would benefit from new measurement instruments. Some could be studied from outside of the firms, using public data on model releases, benchmark progress, and especially tracking what tasks AI researchers automate and how quickly. Some evidence, likely more informative but also harder to access, is likely from inside the frontier AI firms (e.g. S_k). Surveys and field experiments within AI firms could examine the division of labor between humans and AI. Finally, economists should develop new incentivized forecasting instruments, such as prediction markets, to predict capabilities explosion and overall AI capabilities. They should also design mechanisms to facilitate transparency and potential pauses if explosive growth seems more plausible and unprepared for.
- With respect to policy, economists should study the consequences of explosive capabilities growth conditional on it occurring. An explosion in broad capabilities in particular could raise larger questions about the distribution of income, the concentration of market power, and government taxation and regulation.

APPENDIX

A SCALE-DEPENDENT ALGORITHMIC PROGRESS (PT)

Since algorithmic progress is usually not directly observable, it is typically estimated as the increase in capabilities not explained by increases in training compute, just as TFP is estimated by the Solow residual. That is, if we know (from a “scaling law”) that in the absence of algorithmic progress some index of AI capabilities Z would track training compute T , it is conventional to propose the model

$$Z = A \cdot T \tag{2}$$

and calculate A as Z/T . Using a model like “our basic model” in Section 2, the resulting time series for A , in combination with a time series for research inputs R and the semi-endogenous AI R&D function, can then be used to generate a prediction for how fast algorithms would advance after the automation of AI R&D.

Gundlach et al. (2025) show conclusively that model (2) is misspecified, because algorithmic progress in AI development to date has been largely large-scale-biased. For an extreme case of large-scale-biased algorithmic progress, consider instead the model

$$Z = \min(A, T) \cdot T. \tag{3}$$

Here, algorithmic improvements proportionally advance AI capabilities as long as there is the training compute to leverage them (i.e., here, as long as $T > A$). If we stopped scaling the compute, however, capabilities would ultimately stagnate (here, at T^2). Likewise, even if automating AI R&D yields a leap in the quality of the algorithms available, under (3) this will not deliver arbitrarily large advances in AI capabilities until we have accumulated the training compute to leverage these algorithms.

Another potential explanation for the Gundlach et al. finding, however, is that we have been developing large-scale-biased algorithms precisely *because* we have been scaling compute. If we faced a long-term compute constraint, or if an AI model were tasked with recursively self-improving on fixed compute, we might develop algorithms that made ever better use of “small” training runs—just as academic labs to some extent do today. That is, rapid RSI may still be feasible, but only via some kind of “directed technical change”.

REFERENCES

- Daron Acemoglu and James A. Robinson. Persistence of power, elites, and institutions. *American Economic Review*, 98(1):267–293, 2008. doi: 10.1257/aer.98.1.267.
- Philippe Aghion, Benjamin F Jones, and Charles I Jones. Artificial intelligence and economic growth. Technical report, National Bureau of Economic Research, 2017.
- Joel Becker, Nate Rush, Tom Cunningham, David Rein, and Khalid Mahamud. We are changing our developer productivity experiment design. METR, 2026. URL <https://metr.org/blog/2026-02-24-uplift-update/>. Published February 24, 2026.
- Nicholas Bloom, Charles I Jones, John Van Reenen, and Michael Webb. Are ideas getting harder to find? *American Economic Review*, 110(4):1104–1144, 2020.
- Nick Bostrom. *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press, Oxford, 2014.

- Erik Brynjolfsson, Avinash Collis, Felix Eggers, Sophia Kazinnik, and David Nguyen. What is generative AI worth? Technical report, Stanford Digital Economy Lab, 2026. URL <https://digitaleconomy.stanford.edu/publication/what-is-generative-ai-worth/>.
- Jack Clark. Import AI 455: AI systems are about to start building themselves. Import AI newsletter, 2026. URL <https://importai.substack.com/p/import-ai-455-automating-ai-research>. Published May 4, 2026.
- Ajeya Cotra. Six milestones for AI automation. Planned Obsolescence, 2026. URL <https://www.planned-obsolence.org/p/six-milestones-for-ai-automation>. Published April 3, 2026.
- Ben Cottier, Robi Rahman, Loredana Fattorini, Nestor Maslej, and David Owen. How much does it cost to train frontier AI models? Epoch AI, 2024. URL <https://epoch.ai/blog/how-much-does-it-cost-to-train-frontier-ai-models>. Published June 3, 2024.
- Tom Cunningham and Manish Shetty. An apple-picking model of AI R&D. tecunningham.github.io, 2026. URL <https://tecunningham.github.io/posts/2026-03-13-apple-picking-ai.html>. Published April 7, 2026.
- Tom Davidson. What a compute-centric framework says about takeoff speeds. Technical report, Open Philanthropy, 2023. URL <https://www.openphilanthropy.org/research/what-a-compute-centric-framework-says-about-takeoff-speeds/>.
- Tom Davidson, Basil Halperin, Thomas Houlden, and Anton Korinek. When does automating AI research produce explosive growth? feedback loops in innovation networks. NBER Working Paper 35155, National Bureau of Economic Research, April 2026. URL <https://www.nber.org/papers/w35155>.
- Jean-Stanislas Denain and Cheryl Wu. Final training runs account for a minority of R&D compute spending. Epoch AI, Gradient Updates, 2026. URL <https://epoch.ai/gradient-updates/r-and-d-vs-training-compute>. Published March 23, 2026.
- AI Epoch. Key trends and figures in machine learning. *Published online at epochai.org. Retrieved from: <https://epochai.org/trends>*, 2026.
- Epoch AI. Epoch capabilities index (ECI). Epoch AI, 2025. URL <https://epoch.ai/eci>. Accessed June 2026.
- Epoch AI. Data on AI companies. Epoch AI, 2026. URL <https://epoch.ai/data/ai-companies>. Accessed June 2026.
- Ege Erdil, Matthew Barnett, and Tamay Besiroglu. How to fully automate software engineering. Mechanize, Inc. blog, May 2025a. URL <https://www.mechanize.work/blog/how-to-fully-automate-software-engineering/>. Accessed: 2026-06-05.
- Ege Erdil, Andrei Potlogea, Tamay Besiroglu, Edu Roldan, Anson Ho, Jaime Sevilla, Matthew Barnett, Matej Vrzla, and Robert Sandler. GATE: An integrated assessment model for AI automation, 2025b. URL <https://arxiv.org/abs/2503.04941>.
- Daniel Eth and Tom Davidson. Will AI R&D automation cause a software intelligence explosion? Forethought Research Report, March 2025. URL <https://www.forethought.org/research/will-ai-r-and-d-automation-cause-a-software-intelligence-explosion>.

- Maryam Farboodi, Andrew Koh, and Anchi Xia. Data-driven automation. NBER Working paper, 2025.
- Marina Favaro and Jack Clark. When AI builds itself. Anthropic, 2026. URL <https://www.anthropic.com/institute/recursive-self-improvement>. Accessed: 2026-06-08.
- Irving John Good. Speculations concerning the first ultraintelligent machine. In Franz L. Alt and Morris Rubinoff, editors, *Advances in Computers*, volume 6, pages 31–88. Elsevier, 1965. doi: 10.1016/S0065-2458(08)60418-0.
- Hans Gundlach, Alex Fogelson, Jayson Lynch, Ana Trisovic, Jonathan Rosenfeld, Anmol Sandhu, and Neil Thompson. On the origin of algorithmic progress in AI, 2025. URL <https://arxiv.org/abs/2511.21622>.
- Anson Ho. The least understood driver of AI progress. Epoch AI, Gradient Updates, 2026. URL <https://epoch.ai/gradient-updates/the-least-understood-driver-of-ai-progress>. Published February 25, 2026.
- Anson Ho, Tamay Besiroglu, Ege Erdil, David Owen, Robi Rahman, Zifan Carl Guo, David Atkinson, Neil Thompson, and Jaime Sevilla. Algorithmic progress in language models, 2024. URL <https://arxiv.org/abs/2403.05812>.
- Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Henigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. Training compute-optimal large language models, 2022. URL <https://arxiv.org/abs/2203.15556>.
- Charles I. Jones. R&d-based models of economic growth. *Journal of Political Economy*, 103(4):759–784, 1995. doi: 10.1086/262002.
- Charles I. Jones. A.i. and our economic future. NBER Working Paper 34779, National Bureau of Economic Research, 2026. Issued January 2026, revised June 2026.
- Thomas Kwa, Ben West, Joel Becker, Amy Deng, Katharyn Garcia, Max Hasin, Sami Jawhar, Megan Kinniment, Nate Rush, Sydney Von Arx, Ryan Bloom, Thomas Broadley, Haoxing Du, Brian Goodrich, Nikola Jurkovic, Luke Harold Miles, Seraphina Nix, Tao Lin, Neev Parikh, David Rein, Lucas Jun Koba Sato, Hjalmar Wijk, Daniel M. Ziegler, Elizabeth Barnes, and Lawrence Chan. Measuring AI ability to complete long tasks, 2025. URL <https://arxiv.org/abs/2503.14499>. METR.
- METR. Forecasting the impacts of AI R&D acceleration: Results of a pilot study. METR, 2025. URL <https://metr.org/blog/2025-08-20-forecasting-impacts-of-ai-acceleration/>. Published August 20, 2025.
- Beren Millidge. Most algorithmic progress is data progress. *beren.io*, 2025. URL <https://www.beren.io/2025-08-02-Most-Algorithmic-Progress-is-Data-Progress/>. Published August 2, 2025.
- Joel Mokyr. *The Gifts of Athena: Historical Origins of the Knowledge Economy*. Princeton University Press, Princeton, NJ, 2002.

- Jacob Morrison, Noah A. Smith, and Emma Strubell. The hidden cost of thinking: Energy use and environmental impact of lms beyond pretraining, 2026. URL <https://arxiv.org/abs/2605.01158>.
- Toby Ord. *The Precipice: Existential Risk and the Future of Humanity*. Hachette Books, New York, 2020.
- Robi Rahman and David Owen. The training compute of notable AI models has been doubling roughly every six months. Epoch AI, Data Insights, 2024. URL <https://epoch.ai/data-insights/compute-trend-post-2010>. Published June 19, 2024.
- Yangjun Ruan, Chris J. Maddison, and Tatsunori Hashimoto. Observational scaling laws and the predictability of language model performance, 2024. URL <https://arxiv.org/abs/2405.10938>.
- Chenglei Si, Diyi Yang, and Tatsunori Hashimoto. Can LLMs generate novel research ideas? a large-scale human study with 100+ NLP researchers. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. URL <https://arxiv.org/abs/2409.04109>.
- Philip Trammell and Anton Korinek. Economic growth under transformative AI. NBER Working Paper 31815, National Bureau of Economic Research, 2025. Originally issued 2023, revised 2026; forthcoming, Annual Review of Economics.
- Dhruv Trehan and Paras Chopra. Why LLMs aren’t scientists yet: Lessons from four autonomous research attempts, 2026. URL <https://arxiv.org/abs/2601.03315>.
- Parker Whitfill and Cheryl Wu. Will compute bottlenecks prevent an intelligence explosion?, 2025. URL <https://arxiv.org/abs/2507.23181>.
- Parker Whitfill, Ben Snodin, and Joel Becker. Forecasting AI time horizon under compute slowdowns, 2025. URL <https://arxiv.org/abs/2511.19492>.
- Josh You. Most of OpenAI’s 2024 compute went to experiments. Epoch AI, Data Insights, 2025. URL <https://epoch.ai/data-insights/openai-compute-spend>. Published October 10, 2025.
- Eliezer Yudkowsky. Levels of organization in general intelligence. In Ben Goertzel and Cassio Pennachin, editors, *Artificial General Intelligence*, pages 389–501. Springer, Berlin, Heidelberg, 2007. doi: 10.1007/978-3-540-68677-4.12.